

PROJECT DOCUMENTATION · 2026



FuelR

AI-Powered Nutrition & Fitness Tracker



PLATFORM	VERSION	STATUS	DEVELOPER
iOS	1.0.0	TestFlight Live	Harsh Sangwan

- React Native
- Expo SDK 56
- Claude AI
- Firebase
- TypeScript
- EAS Build

01 · OVERVIEW

What is FuelR?

FuelR is a full-stack iOS application that combines AI-powered nutrition coaching with intuitive meal tracking. Users can log food by scanning photos, searching by name, or simply chatting with an AI coach — which understands natural language and takes actions on their behalf in real time.

THE PROBLEM**Nutrition tracking is tedious**

Existing apps require manual searches through large databases, making daily logging feel like a chore that most users abandon within a week.

THE SOLUTION**Conversational + visual logging**

FuelR lets users say "log 200g chicken to lunch" or snap a photo of their meal — the AI handles estimation, logging, and feedback instantly.

KEY HIGHLIGHTS

- AI nutrition coach with agentic tool-use loop powered by Claude Opus 4.8
- Date-aware dashboard — view and edit any day's nutrition log
- Food photo analysis via Claude's vision capabilities
- Per-meal breakdown: Breakfast, Lunch, Dinner, Snacks
- Real-time macro tracking: calories, protein, carbs, fat
- Water intake tracking integrated into AI conversation
- Persistent chat history with Firestore session storage
- Firebase Authentication for secure user accounts

02 · FEATURES

Core Features



AI Nutrition Coach

Conversational agent powered by Claude Opus 4.8 with adaptive thinking. Logs food, checks intake, updates water, and answers nutrition questions via natural language.



Food Photo Scan

Take a photo or pick from gallery — Claude's vision analyses the image, identifies the food, and estimates calories and macros with a health score.



Smart Food Search

Type any food or dish name; the AI estimates nutritional values with editable macros, health score, serving size, and dietary tips.



Date-Aware Dashboard

Healthify-style horizontal date strip. Browse and edit any past or future date's log. AI chat syncs to the selected date automatically.



Water Tracking

Track daily water intake with visual progress indicator. Updateable via tap or through the AI chat: "log 3 glasses of water".



Macro Analytics

Daily calorie ring, macro distribution card, per-meal calorie breakdown, and nutrition insight cards with personalised tips.

ADDITIONAL CAPABILITIES

- Chat history screen with all past AI conversations browsable
- Nutrition goals configuration (calorie, protein, carb, fat targets)
- Health score rating (1–5) on every logged food item
- Quick-add chips for common foods on the Add Food screen

→ Dark-themed UI with gradient accents and smooth animations

03 · AI INTEGRATION

Agentic AI System

CLAUDE OPUS 4.8

ADAPTIVE THINKING

TOOL USE LOOP

How the AI Agent Works

The nutrition AI is built as a custom agentic loop using raw `fetch` calls to the Anthropic API (the Node SDK is not used — it imports `node:fs` which crashes React Native's Metro bundler). The agent runs up to 6 iterations per turn, with thinking blocks preserved across the conversation history. Every tool call targets the date selected on the dashboard — past, today, or future.

`get_today_nutrition`

Fetch full log for the target date

`log_food_item`

Add food to a meal on any date

`update_water_intake`

Set water glass count for a date

`estimate_food_nutrition`

Estimate macros for any food name

`analyze_food_image`

Vision analysis of a food photo

`get_nutrition_goals`

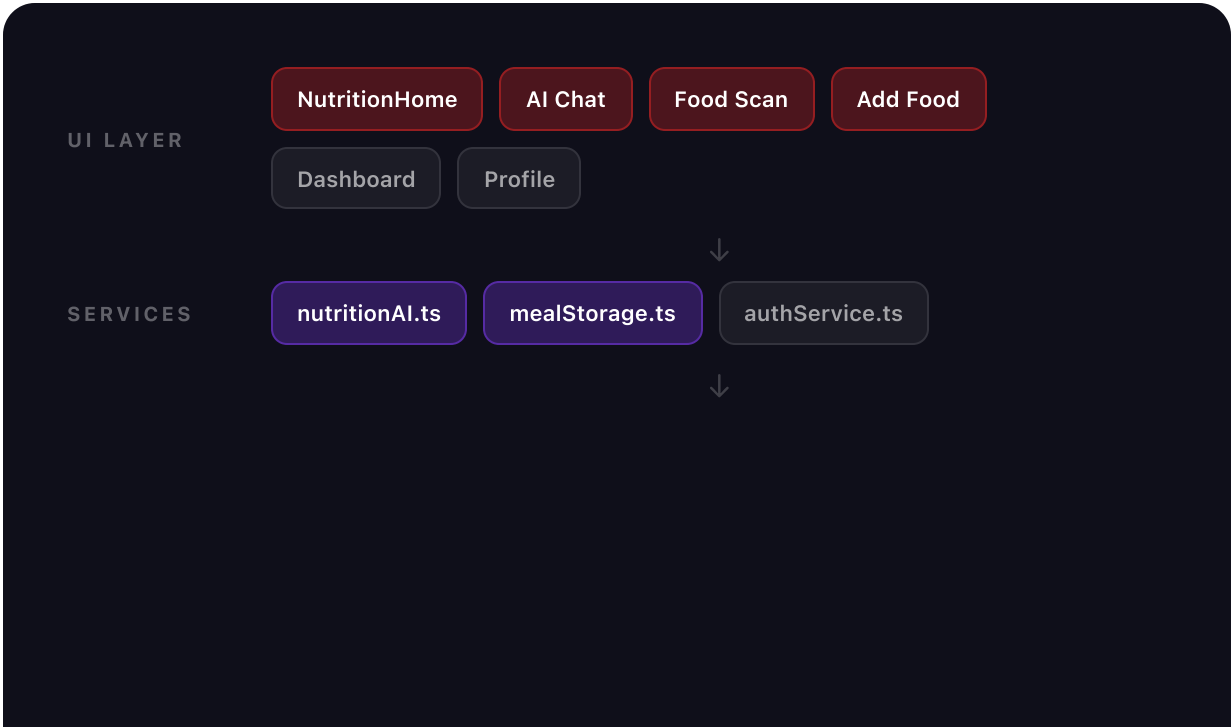
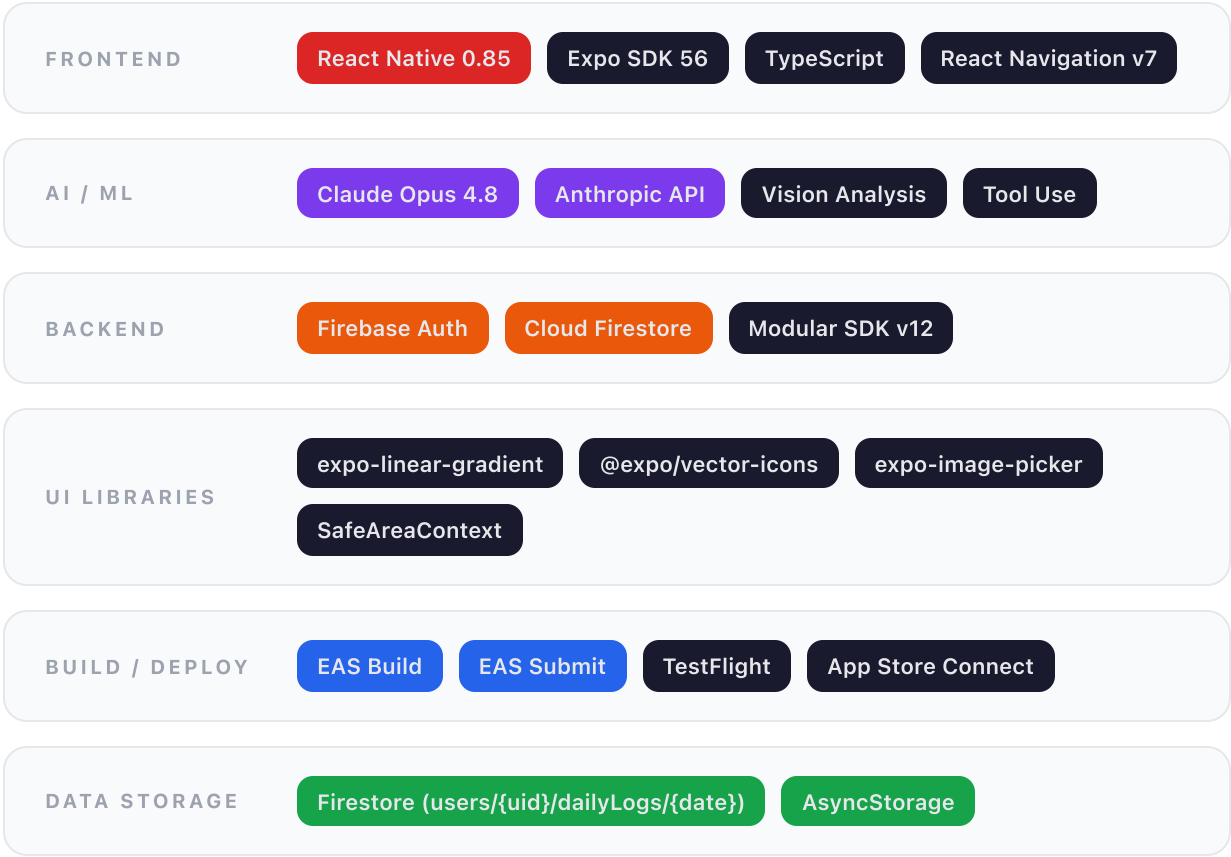
Read the user's calorie/macro goals

DATE SYNCHRONISATION

When the user selects a date on the Nutrition Home screen, that date propagates as a `targetDate` parameter through every navigation action. The AI chat screen reads this parameter, passes it to the agent, and the system prompt explicitly tells Claude which date it is operating on — preventing any confusion between today and historical dates.

04 · TECHNOLOGY

Tech Stack & Architecture



EXTERNAL
APIS

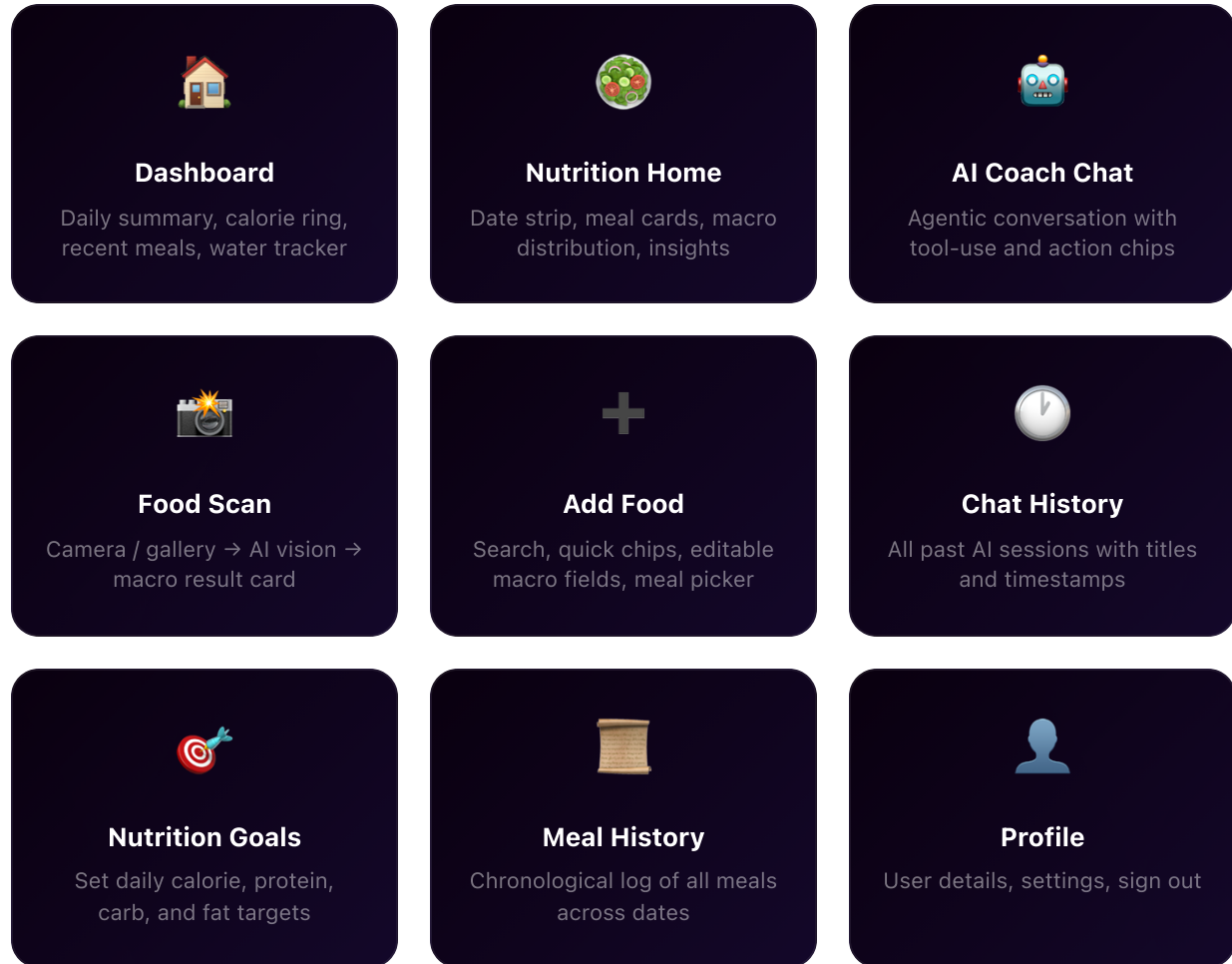
Anthropic API

Cloud Firestore

Firebase Auth

05 · SCREENS

App Screens



06 · DEPLOYMENT

Build & Distribution

✓ Live on TestFlight

App is deployed and accessible to testers via Apple's TestFlight platform. Built and submitted using Expo Application Services (EAS).

⚡ EAS Build Pipeline

Cloud-managed iOS builds with automatic code signing, certificate management, and App Store Connect submission via EAS CLI.

Project: @harsh-sangwan/FuelR

Bundle ID: com.harshsangwan.fuelr

BUILD CONFIGURATION

- Managed Expo workflow — native iOS project generated by `expo prebuild` on EAS servers
- Auto-incrementing build numbers via `eas.json autoIncrement: true`
- Sensitive API keys stored as EAS Secrets — never committed to source
- App Store encryption declaration included (ITSAppUsesNonExemptEncryption: false)

FuelR App

Developed by Harsh Sangwan

Version 1.0.0 · iOS · June 2026

Powered by Claude Opus 4.8 & Firebase